BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 110 · Lesson 110 · Round Robin and Multilevel Feedback Queues

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which scheduler preempts a running task when a shorter one arrives?

Q2. Why can an array provide direct indexed access?

Q3. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

Q4. For the binary value 1010, what is its decimal value?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

time quantum — Maximum continuous CPU interval per RR turn.

round robin — Preemptive cyclic scheduling.

context switch — Saving/restoring CPU state when changing process.

MLFQ — Multilevel feedback queue scheduling.

New learning

Round robin — Ready processes receive a time quantum in cyclic order.

Context switching — Expired slices trigger a switch; very small slices create overhead.

MLFQ — Queues with differing priorities/quanta adapt to process behaviour.

Worked example

RR with quantum 2: P1 burst 5, P2 burst 3; both arrive at t=0. Trace completion.

1. Model answer — P1 runs 0–2, P2 2–4, P1 4–6, P2 6–7 (finishes), P1 7–8 (finishes). Completion times: P2 7, P1 8. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

RR quantum 3: A burst 4, B burst 5; both at time 0. Trace the run order.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

A 0–3 (1 left), B 3–6 (2 left), A 6–7 finishes, B 7–9 finishes. Turnaround A 7 and B 9.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Describe the round-robin algorithm, including the purpose of the time quantum. [6]

Question 2

RR quantum 2: A burst 3, B burst 4, both arrive 0. Produce a Gantt timeline. [6]

Question 3

Explain how MLFQ responds to CPU-bound and interactive tasks and evaluate a very short quantum. [8]

Exit Ticket · Answer before checking

Reducing a time quantum to an extremely small value always improves performance.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Shortest remaining time (SRT).

Q2. Its element locations can be calculated from base address and index.

Q3. Sum 0 and carry-out 1.

Q4. 10.

You Do pages · Independent questions

Q1. FIFO ready queue [2]; each process gets up to a fixed slice [2]; unfinished process goes to back, enabling responsiveness/fairness [2].

Q2. A 0–2, B 2–4, A 4–5, B 5–7 [4]; A completes at 5, B at 7 [2].

Q3. Queues/priorities [2]; jobs move based on behaviour [2]; responsiveness for short/interactive bursts [2]; frequent context switching overhead and trade-offs [2].

Exit Ticket · Model answer

False. Context-switch overhead can dominate.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: